

SIBAM- Sign Inclusive Broken Array Multiplier Design for Error Tolerant Applications

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Abstract—Approximate computing has emerged as a promising technique to develop energy efficient design solutions for error-tolerant applications. Many research efforts have been directed towards proposing approximations in power-hungry multiplier circuits. In this brief, we have introduced two variants of a broken array approximate booth multiplier design called SIBAM with partial error correction through discarded sign bit addition. Experimental results show that up to 63% of energy savings can be achieved compared to accurate multipliers with Mean Relative Error Distance (MRED) constrained at 1.5%. Moreover, extensive analysis shows that the proposed design outperforms state-of-the-art multipliers with energy savings achievable up to 24% at 0.3% MRED constraint.

I. INTRODUCTION

IN recent years, approximate computing has evolved as an efficient solution for developing low power and high-speed digital circuit designs in error-tolerant applications [1]. One of the basic ideas behind approximate computing is to exploit the inherent error resilience of various applications by introducing approximations within functional computations. Arithmetic circuits like adders and multipliers are usually the most power-hungry sources in any digital system. Hence, significant grounds in the research field have been covered in designing approximate arithmetic circuits in the past few years [2], [3].

Multiplier is an integral part of most digital applications and has been identified as a significant source of power consumption. Several research efforts have been dedicated recently in developing approximate multipliers. Examples of approximations involve inaccurate logic functionality, approximate radix encoding techniques, pruning, and rounding of partial products [4]–[10]. One approximation technique is to break the commonly known array multiplier by introducing horizontal or vertical breaks [10], [11]. Employing approximate logic functionality is a common approach where logic expressions are simplified through manipulation of karnaugh map entries. In [4], an approximate 2×2 multiplier is proposed, which is used as a building block to form multipliers with higher bit widths. Approximate compressors are proposed in the literature to facilitate high speed and low power multiplier architectures [7], [8]. Approximate adders proposed in [5] provide faster partial product addition accompanied with partial error recovery. In the pruning technique, either the logic gates computing the partial product accumulation/ addition are eliminated, or the partial products itself are not generated or

utilized in producing the final output. Such examples include Broken Array Multiplier proposed in [10] which operates by pruning carry-save adders depending on horizontal and vertical break level. Approximation in booth multiplier is carried out in [11] by discarding partial products to design power-efficient Broken Booth Multiplier (BBM) by introducing both horizontal and vertical break. A pruning technique known as partial product perforation is proposed in [12] where some of the partial products are omitted while designing a modified booth multiplier. The number of partial products produced can be substantially reduced by employing approximate radix encoding techniques leading to simpler circuit topology. In [6], approximate radix-4 encoding is proposed, which simplifies the partial product generation circuit. Approximate adders are employed in designing a radix-8 encoded approximate multiplier in [13]. Hybrid high radix- 2^k encoding proposed in [14] outperforms the existing approximate radix encoding techniques in terms of energy savings. In this method, high radix partial products are approximated to their nearest power of two. In rounding methodology, some or all partial products are approximated or rounded to an inaccurate partial product resulting in simplified partial product accumulation circuitry [15]. In [16], both partial product perforation and rounding are applied (PR). After doing symmetric partial product perforation, error compensation is applied to each partial product by introducing correction terms. The Pareto efficiency of the proposed hybrid technique in [16] is superior as compared with state-of-the-art multiplier circuits [6], [12]–[14]. A further improvement to the hybrid technique is demonstrated in [17] by using asymmetric partial product perforation (ROUP).

It is to be noted that a multiplier output with a smaller bit width is more power-efficient compared to a multiplier output with a higher bit width. The reasoning is simple enough as more number of output bits would require more processing. Hence, between two multipliers having similar error performance, we should always choose the one with smaller output bit width. The Pareto efficient multipliers proposed in [16] (PR) and [17] (ROUP) respectively reduce the bit-width of the multiplier by employing both perforation and rounding. However, an extra XOR gate is required for each partial product, thus increasing both area and power overhead. On the other hand, although the bit width of BBM (Fig.1a) proposed in [11] is less, it incurs a significant amount of errors. However, if we add the discarded sign bits of partial products,

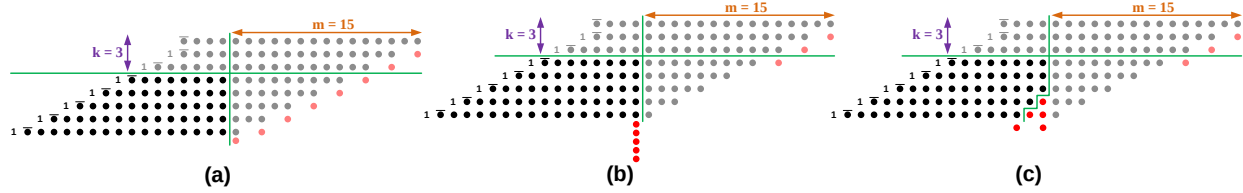


Fig. 1: 16-bit multiplier designs with $k = 3$ and $m = 15$: (a) BBM [11] (b) Proposed SIBAM-SB (c) Proposed SIBAM-AB

the error performance can be significantly increased while undergoing only a small amount of area and power overhead. Hence, in this brief, we propose a new type of broken array multiplier where the sign bits are added to the left most column of partial products. The primary contributions of this brief are as follows:

- 1) We propose two variants of approximate modified booth multiplier design model *SIBAM* targeted towards high speed and energy efficient error-tolerant applications.
- 2) An extensive treatment of the error-energy analysis has been carried out for each of the proposed design models. Subsequently, the designs are compared with the error-energy curves of state-of-the-art multiplier designs.
- 3) Lastly, the proposed multipliers are implemented in two applications: image sharpening and handwritten digit recognition, which showcases the efficacy of the designs at the application level.

This rest of the brief is structured as follows. Section II introduces two variants of proposed multiplier design style *SIBAM*. The experimental results and the subsequent analysis are illustrated in Section III. Section IV concludes the paper.

II. PROPOSED APPROXIMATE MULTIPLIER: SIBAM

The product of two n -bit signed binary numbers X and Y using modified booth encoding is represented in (1).

$$X \times Y = \sum_{i=0}^{\frac{n}{2}-1} (p_i \times 4^i) \quad (1)$$

We obtain our proposed multiplier design by following two steps discussed below:

- **Introducing Horizontal and Vertical Break** At first, we apply partial product perforation technique which is similar to introducing a horizontal break level. As a result, all the rows of partial products above the horizontal break level are discarded. The horizontal break level is characterized by the parameter k . The resultant approximate multiplier equation is represented by (2). $p_i = p_{i,n-1}, p_{i,n-2}, \dots, p_{i,1}, p_{i,0}$ represents all the partial products of the multiplier.

$$(X \times Y)_k = \sum_{i=k}^{\frac{n}{2}-1} (p_i \times 4^i) \quad (2)$$

A vertical break characterized by the parameter m is applied next. Here, all partial products of each row to the right of the vertical break are discarded [Ref Fig. 1a].

The parameter m represents the total number of partial products discarded for 0^{th} partial product row due to application of vertical break only. Let α_i represent the vertical break for i^{th} partial product. Thus, by definition, $m = \alpha_0$. Then the resulting approximate multiplier equation can be written as (3).

$$(X \times Y)_{k,m} = \sum_{i=k}^{\frac{n}{2}-1} \sum_{j=\alpha_i}^{n-1} (p_{i,j} \times 2^j \times 4^i) - \sum_{i=k}^{\frac{n}{2}-1} (p_{i,n} \times 2^n \times 4^i) \quad (3)$$

The applied vertical break can be either symmetrical or asymmetrical as shown in Fig. 1a and Fig. 1b respectively. For symmetrical break, we can write $\alpha_{i+1} = \alpha_i - 2$. Thus, α_i equals to $m - 2i$ for symmetrical break. For $i > \frac{m}{2}$, $\alpha_i = 0$.

- **Sign bit addition** In this step, partial error correction is achieved through addition of discarded sign bits albeit incurring minor area and power overhead. We add the sign bits of each i^{th} non-perforated partial product to the corresponding α_i^{th} bit position. For symmetrical break, all the sign bits are added to α_0^{th} bit position of partial product p_0 . If s_i represents the sign bit of i^{th} partial product, then the approximate multiplication equation for symmetrical vertical break after the addition of sign bits can be represented using (4).

$$(X \times Y)_{k,m} = \sum_{i=k}^{\frac{n}{2}-1} (2^m s_i + \sum_{j=m-2i}^{n-1} (p_{i,j} \times 2^{2i+j})) - \sum_{i=k}^{\frac{n}{2}-1} (p_{i,n} \times 2^{2i+n}) \quad (4)$$

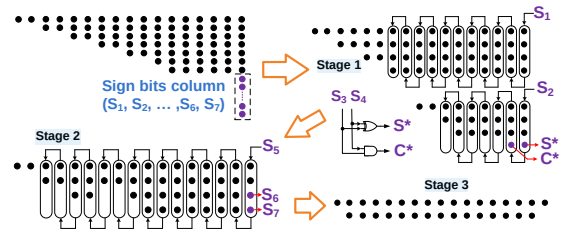


Fig. 2: Partial Product Compression Stages for 16-bit SIBAM-SB Multiplier for $k = 1$ and $m = 15$

When a symmetrical vertical break is applied, the approximate multiplier is termed as *SIBAM-SB*, where *SIBAM* stands for “Sign inclusive Broken Array Multiplier” and *SB* stands for “Symmetrical Break”. Fig. 1(b) and Fig. 1(c) respectively shows 16-bit multiplier examples of proposed designs *SIBAM-SB* and *SIBAM-AB*. The configuration parameters k and m for the specified example are 3 and 15, respectively. The partial products in *SIBAM-SB* are accumulated through either

4 : 2 or 3 : 2 compressor. In *SIBAM-SB*, the number of horizontal rows increases due to addition of extra sign bits. However, these sign bits can easily be accommodated either as an unused carry-in bit of 4 : 2 compressor or can be compressed or added in subsequent accumulation stages. For example, various compression stages of a 16-bit *SIBAM-SB* multiplier are shown in Fig. 2 for $k = 1$ and $m = 15$. The sign bits corresponding to partial product p_i is represented as S_i . Since $k = 1$, the partial product p_0 and corresponding sign bit s_0 is discarded. The sign bits s_1 and s_2 are added as a carry-input to the lowest significant 4 : 2 compressor in stage-1, whereas the sign bit s_5 acts as carry-input for 4 : 2 compressor in stage-2. The remaining sign bits are added as inputs to the 4 : 2 compressors at various stages of compression. To do away with the extra partial product rows, we introduce asymmetrical vertical break at the expense of slightly decreased error performance for the same values of k and m . We have termed it as *SIBAM-AB*, where *AB* stands for “Asymmetrical Break”. The number of partial product rows is reduced through asymmetrical break formation while making sure the inclusion of all sign bits. In our design, we have introduced the asymmetrical break in such a way that the total number of partial product rows is always $(\frac{n}{2} - k + 1)$ in correspondence with accurate multiplier where the number of partial product rows are $(\frac{n}{2} + 1)$. The compression stages of *SIBAM-AB* is simpler than *SIBAM-SB* due to the absence of extra rows which is possible because of the asymmetrical break of the partial products. Compared to BBM, the critical path delay of both *SIBAM-SB* and *SIBAM-AB* remains the same. The delay of both the proposed designs improve in comparison with designs PR or ROUP due to the absence of extra XOR gate delay required for rounding operation.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Error Analysis - *SIBAM*

The error due to application of vertical break to any partial product P_i without sign bit addition is represented using (5). This is the error incurred in a BBM because of vertical break.

$$e_{vb}(i)|_{BBM} = 4^i \left[\sum_{j=0}^{\alpha_i-1} 2^j p_{i,j} + s_i \right] \quad (5)$$

After sign bit addition to the LSB i.e m^{th} bit of multiplier, the error induced to any partial product p_i due to vertical break is given by (6). Here, m^* is equal to m for symmetrical break, whereas m^* varies for asymmetrical break.

$$e_{vb}(i)|_{SIBAM} = 4^i \left[\sum_{j=0}^{\alpha_i-1} 2^j p_{i,j} + s_i \right] - 2^{m^*} s_i \quad (6)$$

Mean Relative Error Distance (MRED) is a significant error parameter used to characterize the error performance of multipliers. It is defined as the average of the cumulative relative error distance over all possible input combinations. MRED of proposed approximate multiplier is represented in (7). $P(X)$ and $P(Y)$ are probability density functions of inputs

X and Y respectively both of which is equal to $\frac{1}{2^n}$ for uniform distribution of inputs.

$$MRED|_{SIBAM} = \sum_{\forall X,Y} P(X)P(Y) \frac{|X \times Y - (X \times Y)_{SIBAM}|}{|X \times Y|} \quad (7)$$

The effect of the addition of discarded sign bits has not been explored in previous works. In this section, we demonstrate how the inclusion of the sign bits significantly improves the error performance. The MRED of *SIBAM-SB* and *SIBAM-AB* is plotted in Fig. 3 for different values of m and k . The MRED of the multiplier without sign bit addition is also plotted. It can be seen there is a substantial improvement in the MRED due to the addition of sign bits for $m \geq 12$ in a 16-bit multiplier. We get similar error improvements for 8-bit and 32-bit multipliers. We have also plotted the error performance of *ROUP* multiplier for similarly defined k and m values. It can be observed that for smaller values of m , the error performance of both *SIBAM-SB* and *SIBAM-AB* is similar to *ROUP*. The error characteristics of *ROUP* show slight improvement at higher values of m .

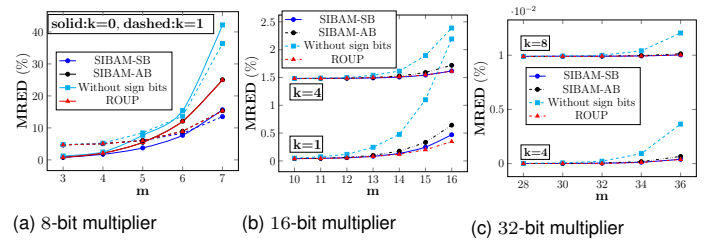


Fig. 3: MRED comparison of multipliers for various values of m

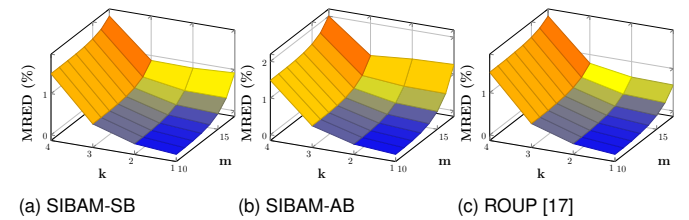


Fig. 4: Variation of MRED in 16-bit multipliers with horizontal break parameter k and vertical break parameter m

The MRED % variation of 16-bit *SIBAM-SB* and *SIBAM-AB* multipliers with horizontal break parameter k and vertical break parameter m is shown in Fig. 4(a) and Fig. 4(b) respectively. It can be observed that the horizontal break parameter k introduces more approximation error compared to vertical break parameter m due to more significance of partial products perforated. Moreover, the MRED % variation with m in *SIBAM-AB* is more as compared with *SIBAM-SB*, since asymmetrical break introduces more errors in a bid to reduce the number of partial product rows. The error characteristics of *ROUP* improves with increasing value of m as compared to *SIBAM-SB* and *SIBAM-AB*.

B. Hardware Usage Analysis

We implemented both 16-bit *SIBAM-SB* and *SIBAM-AB* multiplier designs and experimentally evaluated them in terms

of power, area and delay with state-of-the-art multipliers. Verilog-HDL is used for designing the multipliers which are then synthesized with Synopsys Design Compiler using 90 nm UMC library. The simulations are performed under two conditions: one at nominal frequency of 50 MHz and other at best case frequency achievable specific to each circuit. Two state-of-the-art multipliers are considered for pareto analysis : PR [16] and ROUP [17].

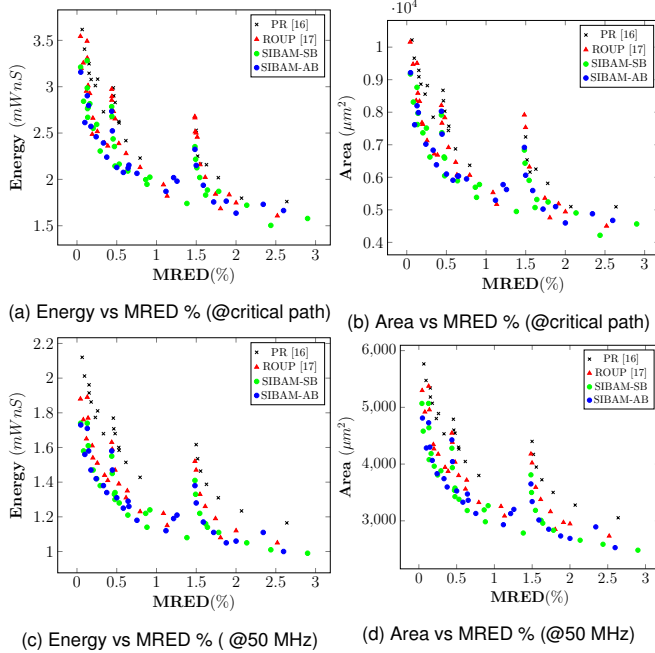


Fig. 5: Pareto analysis of 16-bit SIBAM-SB, SIBAM-AB, PR [16] and ROUP [17] multipliers for various configurations

Fig. 5 demonstrates Pareto analysis of 16-bit *SIBAM-SB* and *SIBAM-AB* multipliers showing energy-error and area-error plots at both nominal clock frequency (50 Hz) and best-case achievable frequency. For comparison, energy-error and area-error plots for various 16-bit PR and ROUP multiplier designs are also shown. It can be observed that *SIBAM-SB* and *SIBAM-AB* both perform better than PR in terms of energy and area, specially at lower MRED range. To understand the results further, we have plotted the energy consumption of approximate multipliers at a specific MRED constraint in Fig. 6. For any MRED constraint, the best optimal result for each design is chosen for the comparison. From Fig. 6, it is apparent that at lower error constraints ($\leq 0.8\%$), *SIBAM-SB* and *SIBAM-AB* show better energy and area savings as compared to ROUP. Moreover, *SIBAM-AB* outperforms both *SIBAM-SB* and ROUP for MRED ranging from 0.6% to 0.8%. Similarly, if we consider the MRED variation from 1.3% to 1.9%, *SIBAM-SB* appears to be the best optimal solution. However, for higher error constraints i.e for higher values of parameter m , the performance of both SIBAM multipliers and ROUP becomes comparable. This observation directly conforms with the error analysis plot shown in Fig. 3, where the error performance of ROUP increases gradually as vertical break parameter m increases. The possible reason is that at higher values of m , the

number of partial products in ROUP gets highly reduced as a result of which the number of extra XOR gates also decreases.

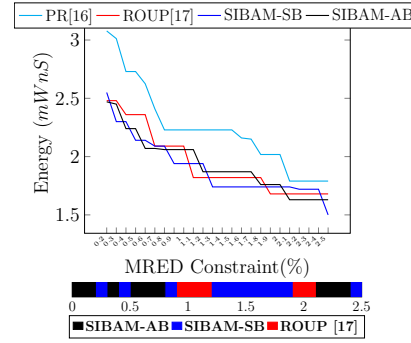


Fig. 6: Energy vs MRED % constraint for various designs

Energy savings of *SIBAM-SB* and *SIBAM-AB* with respect to accurate multiplier, approximate Radix-4 booth multiplier (R4ABM) [6], PR, and ROUP designs at various MRED % constraints are plotted in Fig. 7a, and Fig. 7b respectively. At 2.5% MRED constraint, *SIBAM-SB* and *SIBAM-AB* have 68% and 65% energy savings compared to an accurate multiplier, respectively. *SIBAM-SB* shows 18% and 9% energy savings compared to PR and ROUP at 0.6% MRED constraint. Similarly, at the same error constraint, the respective energy savings of *SIBAM-AB* compared to PR and ROUP are 21% and 12%. As observed in Fig. 6, at higher MRED constraints, the performance of ROUP becomes comparable to *SIBAM-AB* and *SIBAM-SB* multiplier. At low MRED constraints, *SIBAM-AB* performs better than *SIBAM-SB*. As an example, at 0.6% MRED constraint, *SIBAM-AB* shows better energy savings compared to *SIBAM-SB*. At higher MRED % constraints, *SIBAM-SB* tends to perform better than *SIBAM-AB*, which occurs due to comparatively higher deterioration of error performance of *SIBAM-AB*, as observed in Fig. 3 at higher values of vertical break parameter m .

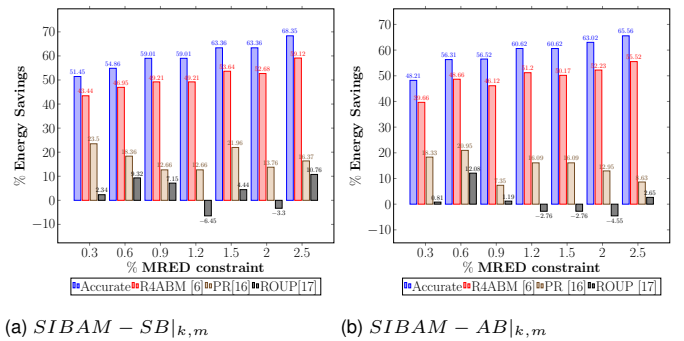


Fig. 7: Energy Savings in 16-bit SIBAM Multiplier under MRED % constraint

C. Application level evaluation

We have evaluated our designed multipliers in two different applications : Image Sharpening and Handwritten Digit Recognition using Convolutional Neural Network (CNN). The

results are also compared with approximate multipliers PR and ROUP.

a) Image Sharpening: The efficiency of the proposed multipliers *SIBAM-SB* and *SIBAM-AB* at application level is demonstrated using image sharpening algorithm. The input image taken is a standard 512×512 16-bit Lena image on to which image sharpening is applied by convolving the image with 5×5 smoothing filter. For evaluation, all accurate multipliers are replaced with approximate versions of the multipliers. The optimal approximate multiplier design (in terms of energy savings) at 0.6% MRED constraint is chosen for comparison. Fig. 8 shows the sharpened images with *SIBAM-SB* and *SIBAM-AB* and their corresponding PSNR values. Among all the designs, the image sharpening filter designed with *SIBAM-SB* shows the best image quality with a PSNR value of 25.81. *SIBAM-AB* has a comparably worst image PSNR value of 18.71. However, at 0.6% MRED constraint, the energy savings of *SIBAM-AB* is also maximum.

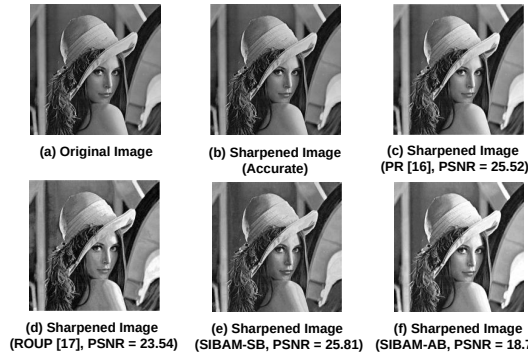


Fig. 8: Image sharpening using approximate multipliers at 0.6% MRED constraint

TABLE I: Handwritten Digit Recognition : Comparison of classification accuracy (in %) for approximate multiplier designs at various MRED (%) constraint

MRED Constraint (%)	Accurate	PR[14]	ROUP[17]	SIBAM-SB	SIBAM-AB
0.5	98.85	98.3	98.1	98.3	98.32
1		98.28	97.82	98.26	98.3
1.5		98.28	97.86	98.2	97.80
2		97.98	97.46	98.2	97.98
2.5		97.92	97.46	97.90	97.80

b) Handwritten Digit Recognition: One of the most popular problems in pattern recognition is handwritten digit recognition [18]. Here, we applied CNN as a classifier on the MNIST data set [19]. The size of the input pixels is 28×28 . The CNN consist of two convolutional layers, each followed by two 22 mean pooling layer. In the first convolutional layer, there are six feature maps connected to the input layer through six 55 kernels. There are 12 feature maps in the second convolutional layer connected through twelve 55 kernels. The final layer consists of 10 output units. Classification accuracy of 98.85% is obtained in a trained network implemented with 16-bit accurate multiplication operations. For evaluation of the proposed multipliers, the 16-bit accurate multiplication operation is replaced with corresponding 16-bit approximate multiplication. The classification accuracy of *SIBAM-*

SB multipliers ranges from 97.9% to 98.3% for MRED % variation from 2.5 to 0.5. Similarly, the accuracy for *SIBAM-AB* multipliers varies from 97.8% to 98.32% (Table I).

IV. CONCLUSIONS

In this brief, two approximate multipliers *SIBAM-SB* and *SIBAM-AB* have been introduced. Design space exploration of both the designs for different configuration parameters is illustrated. Extensive error analysis of the proposed multipliers has been performed. Experimental results show that the proposed multipliers provide upto 24% energy savings than state-of-the-art multipliers at 0.3% MRED constraint.

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